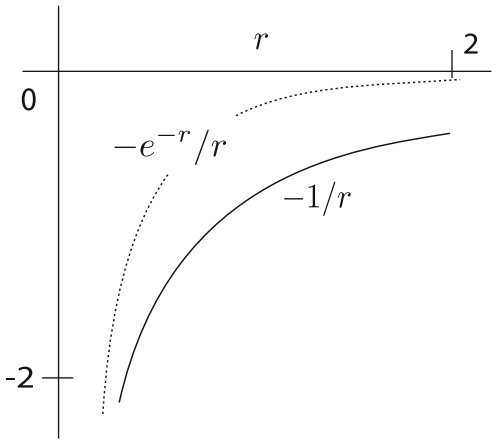


Fig. 1.8 Potentials for Coulombic attraction ($-1/r$, *solid curve*) and screened Coulomb ($-e^{-r}/r$, *dotted curve*) interaction are shown

dihedral angle bonds; these model various configurations defined by the chemical bonding of trios or quartets of atoms and are necessary to give correct structures (although 5-atom bond potentials could also, in principle, be introduced, they are generally found to be unnecessary in practice). When a trio of atoms with labels i, j, and k admits a pair of length bonds, say (i, j) and (j, k), then an additional three-body term will need to be incorporated:

$$\varphi_{ijk}^{\text{ang}} = \frac{k_{ijk}^{\text{ang}}}{2}(\theta_{ijk} - \theta_{ijk}^{0})^2,$$

where the angle θ_{ijk} is given in terms of the positions as

$$\theta_{ijk} = \arccos \frac{(\boldsymbol{q}_i - \boldsymbol{q}_j) \cdot (\boldsymbol{q}_j - \boldsymbol{q}_k)}{r_{ij} r_{jk}}.$$

The leading coefficients describe the energy of the bond in relation to the squared angle, with the rest lengths and angles appearing as parameters. Dihedral potentials typically are modelled using a trigonometric potential function, as

$$\varphi_{ijkl}^{\text{dih}} = k_{ijkl}^{\text{dih}}[1 + \cos(n_{ijkl}^{\text{dih}} \eta_{ijkl} - d_{ijkl}^{\text{dih}})].$$

The angle η_{ijkl} is calculated in the obvious way from the position coordinates, while k, n, d are coefficients appropriate to the local bond organization. In the alanine dipeptide, central dihedral angles Φ and Ψ shown in Fig. 1.7 are defined with reference to a particular bond via the two neighboring bonds along the carbon-nitrogen "backbone" of the chain, as indicated in the illustration. It is then discovered that the most common structures of the molecule can be described in terms of just these two angles, despite the fact that the individual atoms of the system vibrate substantially.